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CASE 5

A Solid Foundation for Child Health

Mexico's Piso Firme Program

The Case at a Glance

HEALTH GOAL: To **improve living standards and health**, especially for mothers and children, among vulnerable groups living in high-density, low-income neighborhoods.

STRATEGY: Government program to **replace dirt floors with cement** in marginalized households using publicly provided materials and community labor.

HEALTH IMPACT: Nearly 20 percent fewer parasites, nearly 30 percent less diarrhea, 20 percent less anemia, and improved cognition in children under six. Levels of depression and perceived stress reduced by 12.5 percent and 10.5 percent, respectively, in their mothers. 408 lives saved and 40,748 disability-adjusted life years averted from 2007 to 2013.

WHY IT WORKED: Simple theory of change and highly visible benefit, which facilitated widespread outreach. High-level government commitment and complementary social programs. State and federal funding. Robust evaluation and culture of learning.

FINANCING: Coahuila State's program budget was US\$5.5 million (US\$162 per household). The federal government spent US\$1.27 billion from 2007 to 2012 (US\$468 per household), in 2010 dollars.

SCALE: Coahuila State installed 34,000 cement floors (2000–2005). The federal government installed 2.36 million cement floors (2007–2013).

All children have diarrhea from time to time, but children who live in houses with dirt floors have it often. When young children crawl and play on dirt floors, they are likely to ingest fecal material, exposing them to the harmful worms and parasites that thrive in excrement.¹ These pathogens can cause persistent diarrhea, among many other ailments. Not only is this unpleasant, but it also robs children of the nutrients that their growing young bodies need and slows their development, leading to wasting, anemia, and cognitive delays.² At its worst, diarrhea depletes the water and salt reserves that everyone needs to survive. Worldwide, diarrhea kills around 750,000 children every year.³

One simple measure to combat diarrhea was called a “no-brainer” by one global health expert: **replace dirt floors in poor households with concrete**. In 2006, an evaluation of the Piso Firme (“Solid Floor”) program in the Mexican state of Coahuila provided irrefutable evidence that flooring improvements offered health and wellness

benefits, particularly for young children and for women.⁴ Living on new floors felt more comfortable and temperate than living on dirt floors. Mothers also liked that they were easier to clean; the new floors actually made them measurably happier.

Piso Firme proved so successful that President Felipe Calderon made a national commitment to eliminate dirt floors in target areas by 2012. Coverage of *pisos firmes* is not yet universal, but the Mexican government is quickly approaching its goal, eliminating an important pathway for diarrhea-causing pathogens that can kill children.⁵

The Toll of Dirt Flooring

Housing is a recognized social determinant of health. Both urban and rural areas have poor housing stock that exposes residents to various types of health risks. Key housing-related health risks include respiratory and car-

This case was originally authored by Miriam Temin.

diovascular diseases from indoor air pollution, communicable disease, injury, and threats to mental health. Urbanization intensifies this close relationship between housing and health.⁶ With rapid urbanization, an ever-growing number of children are spending their formative years in makeshift housing on city peripheries, with governments struggling to catch up and provide adequate housing. The World Health Organization has reported that nearly 40 percent of urban growth worldwide is in unhealthy slum-type housing.⁷ It is in such areas of Mexico that Piso Firme has had its greatest impact.

Dirt floors pose specific risks because they soak up fecal material that leaks or is tracked in by people and animals—a common occurrence in dense urban areas with underdeveloped sewage systems and limited sanitary facilities. Dirt floors are hard to clean and can harbor hookworm, sand fleas, and other worms and protozoans.⁸ In Mexico, dirt floors are associated with an increased burden of intestinal parasites, even when compared with households whose residents are just as poor but have solid flooring, and controlling for other risk factors. Findings from elsewhere in Latin America tell a similar story.⁹

Infestations of worms and parasites pose immediate and long-term risks to children's health and development. Parasitic infestations bring diarrhea and micronutrient malnutrition, which in turn lead to anemia, protein-energy malnutrition, and swelling of the liver and spleen. Worms and parasites affect children's cognitive development, with knock-on effects for their future productivity.¹⁰ As of 2015 there are still no cost-effective drugs for mass prevention or treatment of parasitic protozoan infections. (Deworming drugs have been developed and are in wide use. See chapter 6 in this volume.)¹¹

Worldwide, children have seen significant health gains over the past two decades. In 2010, two million fewer children under five died than in 2000, and the reduction in diarrhea was among the top three reasons for this decline.¹² Consistent with the global trend, Mexican children are healthier than their parents were a generation ago. In the years leading up to the Millennium Development Goal deadline of 2015, Mexico was on track to meet the under-five mortality target: the mortality rate dropped from 20.4 deaths per 1,000 births in 2004 to 16.8 in 2010.¹³

But that is still not good enough. Worldwide, preventable diarrhea kills thousands of children every day—750,000 in 2010. It is the second leading cause of under-five mortality after the neonatal period.¹⁴ And in Mexico, progress on child health is not equitably distributed, and serious health disparities persist. For example,

infant mortality rates are significantly higher in states in the south and southeast than those in the north. In addition, children are more likely to have a fever or diarrhea if they belong to an indigenous ethnic group, if they were born to mothers with a low education level, or if they live in rural areas—and these children are also less likely to receive proper medical treatment.¹⁵ Urban children have a health advantage despite the housing-related health risks in dense informal settlements.

Putting an Idea in Motion: Mexico Moves from Concept to Concrete

In 2000, Enrique Martínez y Martínez was the Revolutionary Institutional Party candidate for governor in the northern state of Coahuila. When his campaign team told him of the many people who were living on dirt floors, he made a promise to the residents of his state to eliminate dirt floors statewide if he became governor.¹⁶ Once elected, Martínez y Martínez launched his signature cement floor installation program, dubbed “Piso Firme,” which provided more than 34,000 homes with much-needed cement flooring upgrades by 2005.¹⁷

The success in Coahuila, coupled with the governor's strong influence at the federal level, piqued national interest in the program and prompted the federal government to make large-scale cement floor installation a national strategic goal.¹⁸ A number of social programs were already under way to combat poverty and marginalization: Mexico's pioneering social protection program, Progresa, later renamed Oportunidades, was the first large-scale conditional cash transfer program of its kind.¹⁹ Simultaneously the government launched Seguro Popular (“Popular Insurance”), a public health insurance program for people living in poverty, a major step on the road to universal health coverage for all Mexicans. Smaller social programs complemented the objectives of Piso Firme, including national deworming activities and programs to improve rural and urban living conditions. Thus, Mexico's federal government was receptive to expanding Piso Firme.

Piso Firme in Action

Piso Firme's simplicity made it an attractive program. A caravan of cement trucks rolled up to eligible communities and delivered US\$150 worth of cement to each participating household, enough to cover about 540 square feet with a five-inch-thick slab. Beneficiaries also received

installation instructions, and family and community members provided the labor. The entire process of mixing and pouring the cement could be completed in just one day.

Piso Firme targeted disadvantaged families in marginalized communities. In Coahuila, program staff determined eligibility by knocking on doors, looking for homes with dirt floors, and offering those homeowners new floors—an offer that nearly all of them accepted.²⁰ Later, the national program used information from the census to identify beneficiary communities based on their degree of “socioeconomic marginalization.”²¹ Individuals within targeted communities were eligible if they could prove that they were living on a dirt floor and qualified as low income.

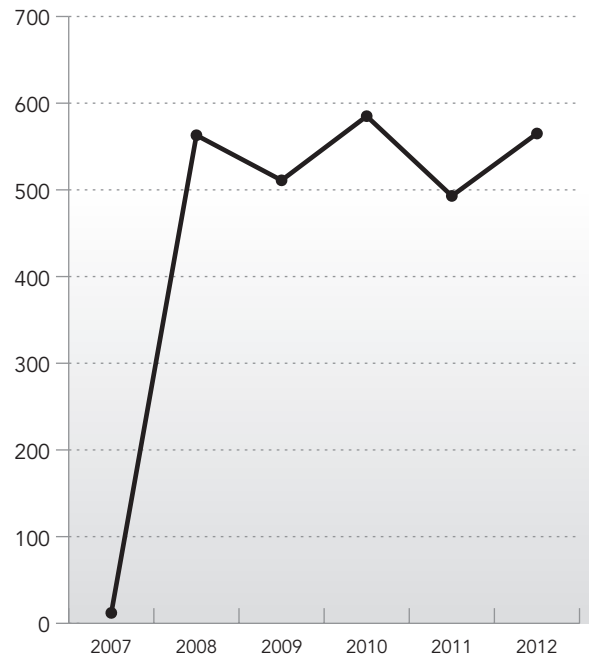
Piso Firme kicked off in Coahuila in 2000. By late 2003, it began its expansion throughout the country via several state and federal poverty-reduction programs.²² By 2005, cement floors had been installed in approximately 300,000 homes. The scale-up of Piso Firme accelerated dramatically from 2007 to 2012, when the total number of cement floors installed reached an impressive 2.7 million (see Figure 1). By 2012, only 4.2 million people (3.6 percent of the population) were still living on dirt floors, down from 8 million (7.3 percent) in 2008.²³

Despite the immense scale of the cement-floor installation, there still are houses with dirt floors, and these houses are not equitably distributed across Mexico.²⁴ Particularly concerning, nearly half of homes in predominantly indigenous areas and more than half in “very highly marginalized” areas have not yet received flooring upgrades.²⁵

The Payoff: Less Diarrhea, Happier Moms

Piso Firme was clearly an operational success in Coahuila, immediately improving the living environment for tens of thousands of residents. Yet Governor Martínez y Martínez and the federal Secretariat for Social Development (Secretaría de Desarrollo Social, or SEDESOL) were not satisfied with an anecdotal demonstration of the program’s success. They wanted to convincingly demonstrate the program’s health impact—especially on children—via an independent impact evaluation, which the state of Coahuila and SEDESOL jointly funded. They brought on board academic and World Bank researchers, including the impact evaluation powerhouse Paul Gertler of the University of California, Berkeley, who had been one of the early evaluators of the federal government’s Progres a program.

Figure 1. Thousands of Cement Floors Installed by Year, Mexico, 2007–2012



Source: CONEVAL (2014).

The impact evaluation yielded impressive findings.²⁶ Two to four years after their families had been offered an improved floor, children under six had nearly 20 percent fewer parasites, a 13 percent lower prevalence of diarrhea, and 20 percent less anemia than children in a matched control community. Less anemia meant better cognitive development: children in the treatment area scored 9 percent higher on the widely used vocabulary test, Test de Vocabulario en Imágenes Peabody (“Picture Peabody Language Development Test”), than children in the control community, whose families did not benefit from the offer of a concrete floor.²⁷

Women also benefited from the flooring upgrades, highlighting the important relationship between mothers’ and children’s health. In the intervention community, the program reduced mothers’ levels of depression and perceived stress by 12.5 percent and 10.5 percent, respectively, compared with the control group. The floors were also associated with mothers’ increased satisfaction with the quality of their housing (15 percent higher than controls) and overall quality of life (19 percent higher).

The impact evaluation was designed to isolate the impact of Piso Firme from other changes in the neighbor-

hood, the immediate living environment, and other potentially confounding factors (see Box 1). This enabled researchers to link the health effect specifically to the role of the cement floors in interrupting parasitic infection, demonstrating that the program had measureable positive impacts well beyond the broader benefits conferred by the economic growth occurring in Mexico at the time.

The Mexican government, using a broad approach to poverty measurement, also reported that the floor upgrades actually reduced poverty.²⁸ The government uses a multidimensional index to measure poverty that combines people's income level with their degree of social deprivation. The index includes people's level of education, social security, health services, access to food,

and quality of dwelling. The quality-of-dwelling metric gauged how solid people's homes were and included, unsurprisingly, the construction material of the floor. This approach reflected the government's strong commitment to cement-floor installation and enabled it to establish the link between flooring upgrades and poverty reduction.

By significantly improving the quality of dwellings across the country, especially in poor and extremely poor rural municipalities—the main target of the flooring upgrades—the scale-up of Piso Firme reduced multidimensional poverty in Mexico. According to calculations by the National Council for the Evaluation of Social Policy (CONEVAL), 9.7 percent of Mexicans would have lived in extreme poverty in the absence of govern-

Box 1. Strength of the Evidence

Measuring the health impact of housing upgrades is methodologically difficult; confounding factors and ethical challenges abound. The Piso Firme evaluation stands out for its robust design, which makes it an important contribution to the slim evidence base on the impact of housing on health in middle-income settings.

The phased rollout of the program created a natural experiment, making it possible to compare Piso Firme beneficiaries with otherwise similar households that had not received the benefit. The evaluators focused their investigation on the La Laguna metropolitan area, an urban agglomeration spanning the border between Coahuila and Durango states. The neighborhoods on either side of the border were socioeconomically and culturally similar, with one important difference: Coahuila was the first state to implement the Piso Firme benefit, starting in year 2000, whereas Durango State still had not done so by that time. Evaluators sampled treated households on the Coahuila side and then used census data from 2000, before program implementation, to identify geographical areas within the Durango neighborhoods that could serve as controls, carefully matching them to treatment areas in Coahuila.³¹

Surveying a randomly selected sample within each group, the researchers were then able to credibly isolate the effects of Piso Firme. They found important health and welfare gains in program areas based on an "intent-to-treat" analysis of the program's effect. The intent-to-treat approach is robust because almost no one turned down a concrete floor when it was offered to them by the program. By including questions about overall housing improvements and the

perceived value of homes with or without concrete floors, the researchers were also able to rule out the possibility that health gains were due to an income effect (for example, because beneficiaries had additional income via the US\$150 in-kind transfer to devote to other health-improving activities). Finally, they exploited the difference in prevalence of concrete flooring across treatment and nontreatment areas to estimate the impact of having a concrete floor on health outcomes. The researchers ultimately found that a higher percentage of concrete flooring in a sample of homes is associated with significant reductions in anemia, diarrhea, and parasite prevalence, whether the floors were installed through Piso Firme or otherwise.

It is important to note limitations related to the Coahuila study design. The authors took steps to ensure comparability across the treatment and control groups; nonetheless, because this is a quasi-experimental study that did not benefit from randomization to treatment, it is possible that residual confounding remained. A replication study is currently under way to shed additional light on the results.³²

The Mexico results are consistent with compelling evidence from elsewhere on the benefits of solid flooring. Un Techo Para Mi País ("A Roof for My Country"), a regional Latin American organization that provides low-cost houses to extremely poor slum dwellers, includes cement flooring in some of its houses. A randomized study of the housing's impact found that the houses improved the occupants' sense of well-being and happiness. And in El Salvador, children in houses with cement floors had less diarrhea than those in the control group.³³

ment-sponsored cement floors; the floors reduced that percentage to 8.9 percent in 2012.²⁹

The researchers who carried out the impact evaluation pointed out the specific characteristics of the children who benefited from Piso Firme in Coahuila. They lived in low-income urban households that were poor, but not very poor. They had adequate nutrition and access to a safe water supply, and they received regular deworming treatment. It is likely that similar children would receive comparable benefits from a flooring upgrade. However, the evaluators caution against generalizing too broadly; where the setting and population differ, the same impact cannot be assured. For example, in rural areas without a safe water supply, children are more likely to contact parasites in a variety of places, which might reduce the benefits of living on a concrete floor.

Questions remain. For example, the health impact of the Coahuila program may have been different depending on which room in the house received the upgrade—covering a kitchen floor may pay bigger health dividends than bedroom upgrades. Similarly, the impact may have varied depending on how much of a household flooring material was changed: an upgrade from all dirt to all concrete floors may offer greater benefits than covering the one remaining dirt floor in a partially upgraded home. Thus it is possible that poorer households benefited more from Piso Firme, given that they were more likely to have all dirt floors prior to the program. A second-generation evaluation replication study is exploring these questions; its findings will yield further insights from Mexico's experience.³⁰

Gains at What Price?

Mexican government support for Piso Firme came from state and federal budgets. In Coahuila State, the total program cost US\$5.5 million from 2000 to 2005, or US\$162 per household. The cost grew significantly under the national expansion: MXN16.3 billion (US\$1.2 billion) from the federal budget for the years 2007 to 2013—an average of US\$460 per household (2007 dollars).³⁴ This difference can be explained in part by the significant increase in transport costs for the national rollout: in Coahuila, the cement was transported in an urban setting, whereas the national effort required trucking cement to more remote areas.³⁵

In Coahuila, researchers found that Piso Firme was cost-effective for cognitive development in particular.³⁶

They measured a 9 percent boost in children's scores on the Picture Peabody Language Development Test after they received a cement floor. This compares favorably with the 12 percent increase that Oportunidades yielded with a unit cost of between US\$210 and US\$750 per household.

To assess the cost-effectiveness of Piso Firme, the total number of diarrhea and anemia cases it averted among the population eligible for a flooring upgrade was modeled for *Millions Saved*. From these two health outcomes, the estimate yielded a mortality impact of 408 averted deaths (all from diarrhea) in the scale-up period (2007 to 2013), which converts to 40,748 disability-adjusted life years (DALYs) averted. An estimate of the cost-effectiveness ratio was not calculated because it was not possible to estimate the full health effects of the program.³⁷

The Keys to Lasting Success

Mexico stands out for its top-notch evaluation system, which has helped the government improve the design of its social programs across the board. It is a legal requirement that all public social programs undergo annual external evaluations, an effort that is led by CONEVAL.³⁸ The government has taken a lead in demonstrating how constructive independent evaluations can influence social policy.³⁹ All evaluation results must be made public, and this transparency helps media and civil society organizations hold the government accountable on its promise to deliver social benefits.⁴⁰

This system enabled the national Piso Firme program to rest on a strong foundation: the association between cement flooring and child health gains that had been established by the impact evaluation. The simplicity of the intervention and evidence of its impact made it easy for planners to describe a theory of change that could be widely understood and widely communicated. The evaluation results emerged around the time that Piso Firme was being scaled up, providing a valuable asset for its supporters. The program's innovative nature plus its rigorous impact evaluation attracted considerable international attention.

Within Mexico, the ability to quantify the health impact of the simple in-kind transfer—the “no-brainer” aspect of the intervention—attracted government funding and sustained political commitment that facilitated scale-up. The leadership of Governor Martínez y Martínez in the first large-scale program and his sway at the

national level, paired with the government's general commitment to social issues and poverty reduction, generated the right environment for widespread expansion. In 2008, Piso Firme received a boost in support and funding when it became part of the federal government's human development strategy, *Vivir Mejor* ("Living Better"). While no longer a discrete program with its own budget, the provision of cement floors is now managed through a number of federal programs aimed at improving living conditions in rural and urban areas.⁴¹ Several states also offer concrete flooring materials as part of their own social development programs.

Cement floor installation provides an excellent opportunity for the government to make its commitment to poverty reduction tangible. The government has capitalized on this, providing placards to homes with upgraded floors reading, "En mi hogar el piso es firme" ("In my home, the floor is solid"), accompanied by a *Vivir Mejor* logo.⁴² These images appear in promotional materials and social media—even President Felipe Calderón tweeted about Piso Firme's success.⁴³

Implications for Global Health

As population growth concentrates in urban areas, policymakers face pressure to confront the health effects of low-quality housing. The success of Piso Firme illustrates the potential of a narrowly defined, well-targeted housing program. Replacing dirt floors with cement ones is likely to improve the health of urban children who already have adequate nutrition and access to a safe water supply—circumstances that leave dirt floors as one of the last routes of parasite transmission. Closing the flooring pathway has relevance for child health, maternal well-being, and urban housing, while also contributing to the burgeoning evidence base on the social determinants of health.

Installing cement floors is not the most effective way to reduce diarrhea in all settings. Basic water, sanitation, and hygiene improvements, among other strategies, may be higher priorities for diarrhea prevention in disease-endemic regions.⁴⁴ And the cost of Piso Firme may be unaffordable for some governments, even as it proves realistic in middle-income settings such as Mexico.

The long-term gains of investments in young children's health have been firmly established, but the other aspect of the Piso Firme story is more surprising: the impact of child health and housing on mothers. Worldwide, women carry the bulk of responsibility for maintaining the family living environment and providing child care, and it is typically their job to fix domestic problems when they arise.⁴⁵ Childhood illness and low-quality housing take a toll on women's mental health and overall well-being; the Piso Firme evaluation results confirmed that improvements in child health and housing can also improve mothers' well-being.

Reports of Piso Firme's impact have had an influence beyond Mexico's borders—similar programs elsewhere often feature the Mexican evaluation results in their promotional materials. Habitat for Humanity has built more than 500,000 houses in 80 countries, often including cement floors in new houses or housing upgrades. Habitat Nicaragua runs a similar program, which includes a campaign to upgrade floors called *Un Piso para Jugar* ("A Floor to Play On").⁴⁶ Project Ethiopia is a grassroots effort started by two Americans to advance development "one village at a time" in Ethiopia. A donation of US\$400 pays for a concrete floor for a household; US\$1,200 provides a floor for a room in a school.⁴⁷

Replacing dirt floors with cement floors seems like a simple and obvious measure, but prior to the launch of Piso Firme, no low- or middle-income country had taken action on this measure at scale. Piso Firme has been a big win for Mexico, one with implications for other middle-income countries with growing slum communities that have begun investing in improving living conditions and housing stock for the poor.⁴⁸ Mexico's experience shows that simple interventions are worthwhile, especially when they are subjected to an impact evaluation and results are published in peer-reviewed journals. Mexico's program provided lessons in targeting, equity, and operational improvements to improve public health. Piso Firme is a valuable contribution to the evidence base on how cement floors and other targeted housing improvements can improve health and well-being for slum-dwelling children and the women who care for them.

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ENDNOTES

1. Ugbomoiko et al. (2007).
2. Hesham, Edariah, and Norhayati (2004).
3. WHO (2013).
4. Cattaneo et al. (2009).
5. CONEVAL (2014).
6. Thomson, Petticrew, and Morrison (2001); WHO Commission on Social Determinants of Health (2008).
7. WHO (2010).
8. Hotez (2008); Collins et al. (2009).
9. Morales-Espinoza et al. (2003); Jacobsen et al. (2007); Lopez et al. (2003); Gamboa et al. (1998).
10. Engle et al. (2007).
11. Harhay, Horton, and Olliaro (2010).
12. Liu et al. (2012).
13. United Nations (2015); INEGI (2013).
14. Liu et al. (2012).
15. Stevens et al. (2008).
16. The government programs Microregiones and Comision Nacional para el Desarrollo de los Pueblos Indigenas included similar interventions prior to 2000. Flooring components of these programs were relatively small and

ENDNOTES, *continued*

- targeted rural areas with a high density of indigenous populations. Microregiones covered selected communities in six municipalities in Coahuila and nine municipalities in Durango in 2003 (see Cattaneo et al. 2009, 4).
17. Cattaneo et al. (2009).
 18. Cattaneo et al. (2009).
 19. A conditional cash transfer program offers a set amount of money to beneficiaries on a predictable schedule when they adhere to established conditions, which are often related to family members' attendance at school and participation in health programs (Levine 2004).
 20. Cattaneo et al. (2009).
 21. CONEVAL (2010).
 22. By 2005, state-level Piso Firme programs were established in Chihuahua (<http://municipios.chihuahua.gob.mx>) and Guanajuato (<http://www.leon.gob.mx>), and the Piso Firme subcomponent of the Programa para Desarrollo Local Microregiones (Microregion Program for Local Development) was expanded (<http://www.sedesol.gob.mx>).
 23. CONEVAL estimates are based on information from the Microregiones Unit of SEDESOL and MCS-ENIGH 2008, 2010, and 2012 (INEGI 2015).
 24. Chávez (2013).
 25. Arroyo (2009).
 26. Cattaneo et al. (2009).
 27. The researchers used an "intent-to-treat" approach for their analysis, as described in Box 1. The same researchers constructed an alternate model to estimate the effect of a complete replacement of dirt floors by cement floors, which they estimated would lead to a 78 percent reduction in parasitic infestations, a 49 percent reduction in diarrhea, an 81 percent reduction in anemia, and a 36–96 percent improvement in cognitive development among young children. Worms were controlled separately through the national albendazole distribution program.
 28. CONEVAL (2014).
 29. CONEVAL (2014).
 30. Basurto et al. (2013).
 31. Cattaneo et al. (2009).
 32. Basurto et al. (2013).
 33. Galiani et al. (2013).
 34. CONEVAL (2014).
 35. Luis Gerardo Mejia Sanchez, personal communication with the author, June 19, 2014.
 36. Cattaneo et al. (2009).
 37. The estimates of program size were calculated from the eligible population. DALYs averted were derived from diarrhea and anemia cases averted and distribution of severity (Velasquez 2004; Lamberti 2012); disability weights (Salomon et al. 2012; Kassebaum et al. 2014); and program efficacy (Cattaneo et al. 2009). The reported large reduction in prevalence of parasites, 19.6 percent, was excluded. Costs were examined, but the estimation of cost-effectiveness in terms of cost per DALY averted was not appropriate and not reported. See Mirelman, Glassman, and Temin (2016).
 38. Székely (2011); Teruel (2013).
 39. Székely (2011).
 40. Pick (2011).
 41. Programa para el Desarrollo de Zonas Prioritarias (PDZP), Vivienda Digna, and Vivienda Rural. Vivienda Digna and Vivienda Rural are part of FONHAPO, the former urban focused and the latter rural focused.
 42. Pick (2011).
 43. Calderon (2012). Piso Firme's visibility has invited some criticism. For example, concerns have been raised that Piso Firme's top-down delivery may override home owners' own priorities for home improvement, such as improving roofing or upgrading sanitation facilities. Irregularities have been reported, from corruption or simulated rigged bids and illegal charges to poor work and delays that are related to elections (see Chávez 2013 ["SEDESOL's 'Piso Firme' Program Failed"]).
 44. Joseph et al. (2006); Ugbomoiko et al. (2007); Center for Sustainable Development (2015).
 45. Langer et al. (2015).
 46. Hábitat para la Humanidad Nicaragua (2015).
 47. For more information, see Project Ethiopia (2015).
 48. WHO and UN-Habitat (2010).